

SRI CHAITANYA IIT ACADEMY, INDIA

AP • TELANGANA • KARNATAKA • TAMILNADU • MAHARASHTRA • DELHI • RANCHI

**PRAGYA G8 – ACHIEVERS | MATHEMATICS
LOGARITHMS | LEVEL 2***Lengthy – correct method, sustained computation.*

$$1. \left[\log_{1/2} \sqrt{\frac{1}{4}} + 6 \log_{1/4} \left(\frac{1}{2} \right) - 2 \log_{1/16} \left(\frac{1}{4} \right) \right] \div \log_{\sqrt{2}} \sqrt[5]{8} =$$

(A) $\frac{5}{2}$

(B) $\frac{18}{5}$

(C) 3

(D) $\frac{6}{5}$

$$2. \left(\log_{\sqrt{5}} 125 \div (\log_5 25)^2 \right) \cdot \left(\log_{1/5} \sqrt{5} \div \log_{0.2} \sqrt[3]{25} \right) =$$

(A) $\frac{3}{2}$

(B) $\frac{9}{8}$

(C) $\frac{3}{4}$

(D) $\frac{8}{9}$

$$3. \log_2 \left(\frac{1}{4\sqrt{4}} \right) + \log_3 \left(\frac{\sqrt[3]{3\sqrt{3}}}{27} \right) + \log_4 \left(\frac{\sqrt[3]{8}}{128\sqrt{2}} \right) =$$

(A) $-\frac{35}{4}$

(B) $-\frac{35}{2}$

(C) $-\frac{21}{4}$

(D) $-\frac{27}{4}$

$$4. \text{ Given } \log_{10} 2 = 0.30103, \text{ the number of digits in } 2000^{2000} \text{ is}$$

(A) 6603

(B) 6602

(C) 6604

(D) 6601

$$5. \text{ Let } N \text{ be the number of digits in } 64^{64}. \text{ Then } N \text{ is (use } \log_{10} 2 = 0.3010)$$

(A) 78

(B) 116

(C) 144

(D) 84

$$6. \text{ If } \log_{ab} a = 4 \text{ and } \log_{ab} \left(\frac{\sqrt[3]{a}}{\sqrt{b}} \right) = \frac{p}{q} \text{ where } p \text{ and } q \text{ are coprime, then } |p - q| \text{ equals}$$

(A) 5

(B) 6

(C) 11

(D) 17

7. If $x = (\text{antilog}_2 3)(\text{antilog}_3 4)$, $y = \text{antilog}_6 2$ and $\frac{x}{y} = \frac{p}{q}$ in lowest terms ($p, q \in \mathbb{N}$), then $p + q$ equals

(A) 17

(B) 18

(C) 20

(D) 19

8. The number of zeros after the decimal point and before the first significant digit in $\left(\frac{5}{6}\right)^{100}$ is

(A) 8

(B) 7

(C) 5

(D) 10

9. $\left(\log_8 27 - \log_{0.5} \frac{1}{3}\right) \cdot \left(\frac{\log_3 12}{\log_{36} 3} - \frac{\log_3 4}{\log_{108} 3}\right) =$

(A) 2

(B) 0

(C) 1

(D) -1

10. $\log_{\sqrt{6}} 3 \cdot \log_3 36 + \log_{\sqrt{3}} 8 \cdot \log_4 81 =$

(A) 16

(B) 4

(C) 20

(D) 12

11. The number of solutions of $\log(2x) = 2 \log(4x - 15)$ is

(A) 4

(B) 3

(C) 2

(D) 1

12. The solution set of the equation $\log(8 - 10x - 12x^2) = 3 \log(2x - 1)$ is

(A) {5}

(B) {1}

(C) $\emptyset$

(D) {3, 2}

13. Suppose $\log_{10}(x - 2) + \log_{10} y = 0$ and $\sqrt{x} + \sqrt{y - 2} = \sqrt{x + y}$. Then the value of $(x + y)$ is

(A) $2\sqrt{2}$ (B) $2 + 2\sqrt{2}$

(C) 2

(D) $4 + 2\sqrt{2}$

14. The real value of x for which $\log_6 9 - \log_9 27 + \log_8 x = \log_{64} x - \log_6 4$ holds is

(A) $\frac{1}{8}$ (B) $\frac{1}{2}$ (C) $\frac{1}{16}$ (D) $\frac{1}{4}$

15. The reals x and y satisfy $\log_8 x + \log_4 y^2 = 5$ and $\log_8 y + \log_4 x^2 = 7$. The value of xy is
(A) 2^{24} (B) 2^9
(C) 2^{12} (D) 2^{18}
16. If $\log_{\sqrt{2}} \sqrt{x} + \log_2 x + \log_4 x^2 + \log_8 x^3 + \log_{16} x^4 = 40$, then x equals
(A) 128 (B) 64
(C) 256 (D) 32
17. Let B, C, P, L be positive reals with $\log(BL) + \log(BP) = 2$, $\log(PL) + \log(PC) = 3$ and $\log(CB) + \log(CL) = 4$ (base 10). Then the product $BCPL$ equals
(A) 10^2 (B) 10^9
(C) 10^4 (D) 10^3
18. Let $u = (\log_2 x)^2 - 6 \log_2 x + 12$ where x is real. Then the equation $x^u = 256$ has
(A) exactly one solution for x
(B) no solution for x
(C) exactly two distinct solutions for x
(D) exactly three distinct solutions for x
19. If $(49)^{3 \log_{\sqrt{343}} \sqrt{x}} - 2x - 3 = 0$, then x equals
(A) -1 (B) 1
(C) 3 (D) 2
20. If $\log_8 a + \log_8 b = (\log_8 a)(\log_8 b)$ and $\log_a b = 3$, then the value of a is
(A) 2 (B) 8
(C) 15 (D) 16
21. Let $N = (2 + 1)(2^2 + 1)(2^4 + 1) \cdots (2^{32} + 1) + 1$. Then $\log_{256} N$ equals
(A) 32 (B) 15
(C) 8 (D) 16
22. If $(21.4)^a = (0.00214)^b = 100$, then the value of $\frac{1}{a} - \frac{1}{b}$ is
(A) 4 (B) 0
(C) 1 (D) 2
23. The reals x and y satisfy $\log_{16} x + \log_4 y^2 = 9$ and $\log_{16} y + \log_4 x^2 = 6$. Then xy equals
(A) 2^{-12} (B) 24
(C) 2^{12} (D) 3^{12}

24. Given $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, the number of digits in 6^{100} is
- (A) 76 (B) 77
(C) 75 (D) 78
25. Given $\log_{10} 2 = 0.3010$ and $\log_{10} 5 = 0.6990$, the number of zeros after the decimal point and before the first significant digit of $10^{100(\log_{10} 2 - \log_{10} 5)}$ is
- (A) 39 (B) 38
(C) 37 (D) 40